

Loops Questions

① Print 1 to N, N from user.

Soln let n = Num(prompt("Enter no.-"))

```
for (let i = 0; i <= N; i++) {  
    console.log(i)  
}
```

② Print N to 1, n from user.

let n = Num(prompt("Enter no.-"))

```
for (let i = n; i >= 1; i--) {  
    console.log(i)  
}
```

③ Print even nos 1 to N.

let n = Num(prompt("Enter no.-"))

```
for (let i = 2; i <= N; i += 2) {  
    console.log(i)  
}
```

④ Print odd nos 1 to N.

let n = Num(prompt("Enter no.-"))

```
for (let i = 1; i <= N; i += 2) {  
    console.log(i)  
}
```

⑤ Print table of n, take n from user:-

let n = Num(prompt("Enter no.-"))

```
for (let i = 1; i <= 10; i++) {  
    console.log(i * n)  
}
```

⑥ Sum 1 to n.

```
let n = Num(prompt("Enter no-"))
```

```
let sum = 0
```

```
for (let i = 1; i <= n; i++) {
```

```
    console.log(i)
```

```
    sum = sum + i
```

```
}
```

```
console.log(sum)
```

⑦ Count even no 1 to n.

```
let n = Num(prompt("Enter no-"))
```

```
let
```

```
let count = 0
```

```
for (let i = 2; i <= n; i += 2) {
```

```
    count++
```

```
}
```

```
console.log(count)
```

~~shi hoi~~
bc
VS code jare
mke

```
let n = Num(prompt("Enter no-"))
```

```
let count = 0;
```

```
for (let i = 1; i <= n; i++) {
```

```
    if (i % 2 == 0) {
```

```
        count++
```

```
    }
```

```
}
```

```
console.log(count)
```

⑧ Sum of even no 1 to n.

```

let n = Num(prompt("Enter no-"))
let sum = 0;
for (let i = 2; i <= n; i += 2) {
    sum += i
}
console.log(sum)

```

⑨ Count nos divisible by 3 from 1 to n.

```

let n = Num(prompt("No-"))
let count = 0;
for (let i = 1; i <= n; i++) {
    if (i % 3 == 0) {
        count++
    }
}
console.log(count)

```

⑪ Count (+)ve, (-)ve, and zero.

```

let n = Num(prompt("How many nos"))
let posi = 0
let nega = 0
let zero = 0
for (let i = 1; i <= n; i++) {
    let num = Num(prompt("Enter no-"))
    if (num > 0) {
        posi++
    } else if (num < 0) {
        nega++
    } else if (num == 0) {
        zero++
    }
}

```

```

console.log("Positive", posi)
console.log("Negative", nega)
console.log("Zero", zero)

```

ab tak jo sabse bada mila, usko largest me rakhho. Har nye no. ko usse compare karo.
if new no. bada hai, then largest ko update kro.

⑩ find the largest number:

if (arr[0] > min) min = arr[0]

for (i = 1; i < arr.length; i++)

{ if (arr[i] > min) min = arr[i]; }

return min;

(arr[0]) galat hai

arr[0] min hai isko update krke arr[0] se bada

number ko min me rakhte hain aur arr[0] ko min

me rakhte hain aur arr[0] ko min me rakhte hain

aur arr[0] ko min me rakhte hain

aur arr[0] ko min me rakhte hain

aur arr[0] ko min me rakhte hain

aur arr[0] ko min me rakhte hain

(arr[0] ko min me rakhte hain)

if (arr[0] > min) min = arr[0]

for (i = 1; i < arr.length; i++)

{ if (arr[i] > min) min = arr[i]; }

return min;

(arr[0] ko min me rakhte hain)

{ if (arr[i] > min) min = arr[i]; }

return min;

(arr[0] ko min me rakhte hain)

return min;

(arr[0] ko min me rakhte hain)

return min;

(arr[0] ko min me rakhte hain)

★
12) Reverse a Number → (573)

Soln
let num = 573
let rev = 0
while (num > 0) {
 let rem = num % 10
 ~~rev~~ rev = rev * 10 + rem
 num = parseInt(num / 10)
}
console.log(rev)
Print

Q4) ~~Count~~ no. of digits in a number: 576

let num = 576
while (num > 0) {
 let rem = num % 10
 console.log(rem)
 num = parseInt(num / 10)
}

// 6
7
5

let num = 576

let str = String(num)

for (let i = 0; i < str.length; i++)

console.log(str[i])

}

// 5
7
6

Q13) Sum of digits →

let num = 573
let sum = 0
while (num > 0) {

let rem = num % 10

sum += rem

num = parseInt(num / 10)

}

console.log(sum) // 15

⑭ Count the no. of digits —

```
let num = 2376
```

```
let count = 0
```

```
while (num > 0) {
```

```
    let rem = num % 10
```

```
    count ++
```

```
    num = parseInt(num / 10)
```

```
}
```

```
console.log(count) // 4
```

⑮ Check Palindrome

```
let num = 8008
```

```
let org = 8008
```

```
let rev = 0
```

```
while (num > 0) {
```

```
    let rem = num % 10
```

```
    rev = rev * 10 + rem
```

```
    num = parseInt(num / 10)
```

```
} → console.log(rev)
```

```
if (org == rev) {
```

```
    console.log(`${org} is a Palindrome`)
```

```
} else {
```

```
    console.log(`${org} is not a Palindrome`)
```

```
}
```

(16) Factorial →

```
let num = 5
```

```
let pro = 1
```

```
for (let i = 1, i <= 5; i++) {
```

```
    pro *= i
```

```
}
```

```
console.log (pro)
```

or

```
let num = 5
```

```
let pro = 1
```

```
for (let i = 5; i > 0; i--) {
```

```
    pro *= i
```

```
}
```

```
console.log (pro)
```

(17) Print Factors →

```
let num = 10
```

```
let
```

```
for (let i = 1; i <= 10; i++) {
```

```
    if (num % i == 0) {
```

```
        console.log (i)
```

```
    }
```

```
}
```

(18) Count Factors →

```
let num = 10 → let count = 0
```

```
for (let i = 1; i <= 10; i++) {
```

```
    if (num % i == 0) {
```

```
        count++
```

```
    }
```

```
}
```

```
console.log (count)
```

(19) Prime no. (prime hai ya nhi)

```
let num = Number(prompt("enter ur no-"))
```

```
let isprime = true
```

```
if (num <= 1) {
```

```
    isprime = false
```

```
} else {
```

```
    for (let i = 2; i < num; i++) {
```

```
        if (num % i == 0) {
```

```
            isprime = false
```

```
            break;
```

```
        }
```

```
    }
```

```
}
```

```
if (isprime) {
```

```
    console.log(`Yes ${num} is a prime no`)
```

```
} else {
```

```
    console.log(`No ${num} is not a prime no`)
```

```
}
```

Example: Maan le 7 ko check karna hai toh hum dekhenge ki 2 se lekar num-1 (7-1=6) 2 se 6 tak 7 ko divide krenge if vo 2-6 me kisi bhi no. se divide ho jata hai toh vo prime no nhi hai.

Q0 Print prime no. from 1 to N:-

```
let num = Num(prompt("Enter Number:-"))
let isprime = true. → wrong kar ek ke bgy true rhega
for (let i = 2; i <= num; i++) {
  let isprime = true
  for (let j = 2; j < i; j++) {
    if (i % j == 0) {
      isprime = false
      break
    }
  }
  if (isprime) {
    console.log(i)
  }
}
```

Q1 → num=5 Outer loop 'i=2' → isprime = true → inner loop 'j=2'
 inner loop condⁿ: $j < i \rightarrow 2 < 2 \rightarrow \text{false}$ " " chla kr nhi.
 → isprime = true ∴ print 2

I2 → i=3 ∴ $i=3 \rightarrow \text{isprime} = \text{true} \rightarrow j=2 \rightarrow i \% j == 0$
print 3 ∴ $3 \% 2 == 0 : \text{false}$

I3 → $i=4 \rightarrow j=2 \rightarrow i \% j == 0 : 4 \% 2 == 0 \text{ true} \therefore \text{isprime} = \text{false}$
 break.

I4 → $i=5 \rightarrow j=2 \rightarrow 5 \% 2 == 0 : \text{false} \therefore \text{isprime} = \text{true}$
 print → 5

prime
 → i → vo num hai jisko hum check kar rhe hai.
 → j → current divisor jisse hum i ko check kr rhe

Q1) Print Fibonacci Series:

① let num = Number(prompt("Enter ur no -"))

② console.log(a) → can't declare yet.

③ let a = 0 } because every fibonacci series starts with
0 & 1

④ let b = 1 } that is why we console a & b firstly.

⑤ console.log(b)

⑥ for (let i = 3; i <= num; i++) {

⑦ let next = a + b;

⑧ console.log(next)

⑨ a = b

⑩ b = next

}

Chainology → let num = 7 ∴ series : 0 1 1 2 3 5 8

a. ② & ③ → ~~let~~ a = 0, b = 1

b. ④ & ⑤ → print : 0, 1

c. ⑥ → loop starts & here "i" fibonacci ki value nhi hai
 ↳ "i" simply count kar sha hai ki ab konsa fibo. no print hoga
 ↳ already 0, 1 printed means 1st & 2nd no ∴ i starts with
 3rd num that's why i = 3.

d. ⑦ next = a + b ∴ 0 + 1 = 1 → 3rd no. of fibo. series.

e. ⑧ console.log(next) ∴ printed ① // 0, 1, 1

f. ** ⑨ & ⑩ → a = b, b = next.
 ↳ a = 1, b = 1

g. i++ means i = 3 → i = 4 now, we will get our 4th no.

currently : a = 1, b = 1 ∴ next = 2 → 4th fibo. no.

↳ now, a = b ∴ a = 1

b = next ∴ b = 2

b. i++ means i = 5, n : a = 1, b = 2 ∴ next = a + b = 3 → 5th fibo. no.

↳ now, a = b ∴ a = 2
 b = next ∴ b = 3

Q22) Find second largest number:-

145

$$\text{now, } 1 + 24 + 120 = 145$$

$$1! = 1$$

$$4! = 24$$

$$5! = 120$$

which is equals
to original no.

Page:	
Date:	

Check

Q236 Find strong no:- (iske har digit ke factorial ka sum = original no)

digits nikalo $\rightarrow$ har digit ka factorial nikalo $\rightarrow$ factorials ke add kro
 $\rightarrow$ original no se compare karo.

Solⁿ let num = Num(prompt("Enter ur no-"))

let original = num

let sum = 0

while (num > 0) {

let rem = num % 10

let pro = 1

for (let i = 2; i <= rem; i++) {

pro *= i

}

sum += pro

num = parseInt(num / 10)

}

console.log('sum of factorials-', sum)

if (sum == original) {

c.log('Yes {original} is a strong no.')

} else {

c.log('No {original} is not a strong no.')

}

Q24 Armstrong Number (no. equal to the sum of its own digits each raised to the power of the total no. of digits)

- 1. no. ka total digits count krna
- 2. digits ^{total digits} ka sum krna
- 3. original no. se compare.

```
=> let num = Num(prompt("Enter ur no-"))
```

```
let original = num; no need
```

```
let temp = num;
```

```
let count = 0;
```

```
// counting total digits:
```

```
while (temp > 0) {
```

```
    count++
```

```
    temp = parseInt(temp/10)
```

```
}
```

```
// Armstrong sum:
```

```
temp = num
```

```
let sum = 0
```

```
while (temp > 0) {
```

```
    let rem = num % 10
```

```
    sum += rem**count
```

```
    temp = parseInt(temp/10)
```

```
}
```

```
// checking...
```

```
if (sum == numoriginal) {
```

```
    console.log('Yes {original} is arm's no')
```

```
} else {
```

```
    console.log('No arm's no')
```

```
}
```

Q255 Perfect Number : (is a +ve whole number that equals the sum of its proper divisors) (like all its +ve divisors excluding the no. itself)

```
let num = Num(prompt("Enter ur no-"))  
let sum = 0
```

```
for (let i = 1; i < num; i++) {  
    if (num % i == 0) {  
        sum += i  
    }  
}
```

```
}
```

```
if (sum == num) {
```

```
    c.log(`{num} is perfect no`)
```

```
} else {
```

```
    c.log(`{num} isn't perfect no`)
```

```
}
```